The Economic Geography of Europe:

Regional Disparities in GDP per Capita

Reproducible empirical study

23 July 2026

Abstract

This paper measures the magnitude and spatial organization of regional economic disparities in Europe using Eurostat's 2023 GDP-per-capita index and NUTS 2024 level-2 boundaries. An exact one-to-one inner join produces 276 regions: 244 in the EU27 and 32 in four candidate countries. Across equally weighted regions, the median index is 84, the Gini coefficient is 0.219, and the 90-to-10 percentile ratio is 2.62. Spatial association is assessed on log GDP per capita using row-standardized first-order queen contiguity. After excluding 21 regions without a contiguity neighbor, global Moran's I is 0.600 ($p = 0.00001$, 99,999 random-label permutations). The result remains large under raw-index, rook, EU-only, largest-component, and nearest-neighbor specifications. Benjamini-Hochberg-adjusted local Moran statistics identify 4 high-high, 16 low-low, and 1 high-low regions. These results establish a pronounced descriptive spatial pattern, not causal spillovers, convergence, or the effects of any particular regional policy.

Keywords: regional inequality; GDP per capita; NUTS 2; spatial autocorrelation; Moran's I; Europe

1 Introduction

Large differences in regional productivity and income generation remain central to European economic policy. The relevant geography is not only national: capital regions, corporate centers, industrial belts, rural peripheries, and border areas can differ sharply within the same country. At the same time, neighboring regions share markets, infrastructure, institutions, and history. Regional inequality and spatial clustering are therefore distinct empirical questions. Dispersion measures describe how unequal regional values are, while spatial statistics ask whether similar values are located near one another.

This paper asks: *How strongly, and where, was 2023 GDP per capita spatially clustered among neighboring NUTS 2 regions with available Eurostat data?* It makes three contributions. First, it constructs an auditable cross-section from two official snapshots and documents the exact coverage created by the requested inner join. Second, it reports region-equal distributional statistics and a complete choropleth with insets for outermost regions. Third, it combines global and local Moran statistics with permutation inference, explicit zero-neighbor treatment, multiple-testing adjustment, and prespecified sensitivity checks.

The evidence is clear but deliberately descriptive. The region-equal mean is 89.6 and the median is 84 relative to an EU27 benchmark of 100. The spatial result is stronger still: Moran's I for log GDP per capita is 0.600, compared with a random-label expectation near zero. The pattern is robust to alternative variable transformations, weights, and samples. Local evidence is more selective: 21 regions are retained after Benjamini-Hochberg adjustment, of which 16 form low-low cores. Nothing in this one-year design identifies why the pattern arose.

2 Literature review

The theoretical starting point is the possibility that economic activity concentrates endogenously. In the core-periphery model of Krugman [1991], increasing returns, transport costs, and market-size feedback can produce agglomeration. That mechanism is a useful interpretation of clustered production, but a cross-sectional map cannot test it. A separate growth literature studies beta and sigma convergence. Sala-i-Martin [1996] distinguishes faster growth among initially poorer regions from declining cross-sectional dispersion. A single 2023 observation tests neither.

European empirical work has long emphasized the spatial distribution rather than only an average convergence coefficient. López-Bazo et al. [1999] connect distributional dynamics and spatial association across EU regions. Le Gallo and Ertur [2003] use global and local spatial statistics to document persistent high- and low-income clusters in 138 European regions over 1980–1995. Ertur et al. [2006] further show that spatial dependence and spatial regimes matter for estimated convergence processes. More recently, Iammarino et al. [2019] synthesize the heterogeneous trajectories behind European regional inequality and caution against one-size-fits-all interpretations. The Ninth Cohesion Report likewise describes EU-wide convergence alongside persistent territorial challenges [European Commission, Directorate-General for Regional and Urban Policy, 2024].

The present study updates the descriptive spatial cross-section without claiming a convergence test. Global spatial association is measured with Moran's I [Moran, 1950]; local cluster cores and spatial outliers are identified with Anselin's local indicators of spatial association [Anselin, 1995]. Because hundreds of local hypotheses are examined, the main map uses the adjustment proposed by Benjamini and Hochberg [1995].

3 Data

3.1 Economic and geographic sources

The economic input is Eurostat dataset `nama_10r_2gdp`, "Gross domestic product at current market prices by NUTS 2 region" [Eurostat, 2026a]. The API query filters 2023 and unit code `PPS_HAB_EU27_2020`; the response records annual frequency. The index expresses purchasing power standard per inhabitant as a percentage of the EU27 average. Its benchmark of 100 is population weighted, not the mean of NUTS 2 observations; an equally weighted regional mean need not equal 100.

The downloaded CSV has 416 unique geographic keys: 31 national values, 108 NUTS 1 values, 276 NUTS 2 values, and the EU27 aggregate. The geographic input is the GISCO NUTS 2024 level-2 region shapefile at 1:20 million resolution in WGS 84 (EPSG:4326) [Eurostat GISCO, 2026]. NUTS 2 denotes "basic regions for regional policies," and the 2024 classification is valid from 1 January 2024 [Eurostat, 2026c]. The raw geometry contains 299 valid, unique level-2 features.

3.2 Cleaning, join, and sample

Identifiers are preserved as strings, including meaningful zeroes, and the economic constants, numeric values, uniqueness, projection, geometry validity, and NUTS level are asserted before joining. A one-to-one inner join of CSV `geo` to shapefile `NUTS_ID` yields 276 regions and no unmatched GDP-side NUTS 2 key. The inner join itself removes every national, NUTS 1, and EU aggregate, exactly as required.

The resulting sample is broader than the EU27: it contains all 244 EU NUTS 2 regions plus 32 regions in Montenegro, North Macedonia, Serbia, and Türkiye. Twenty-three mapped regions have no selected GDP observation: Albania (3), Bosnia and Herzegovina (3), Switzerland (7), Iceland (1), Liechtenstein (1), Norway (7), and Kosovo (1). The estimand is therefore the set of Eurostat NUTS 2 regions with an available value, not every region conventionally understood as Europe.

Eurostat marks 101 joined observations provisional and 1 estimated. These flags are retained. The analysis does not transform the archived source files; byte sizes, SHA-256 hashes, URLs, retrieval date, anti-joins, edge lists, software versions, and every derived output accompany the paper.

3.3 Measurement

GDP per capita measures production located in a region, not disposable household income or welfare. Commuting can separate workplace production from residence, and the regional location of multinational profits can elevate corporate centers [Eurostat, 2025]. Moreover, regional purchasing-power parities do not exist; Eurostat applies national conversion factors within each country [Eurostat, 2026b]. The analysis therefore treats the index as a harmonized production measure and does not attach welfare interpretations to small differences.

4 Methods

4.1 Distributional description

Each NUTS 2 region receives equal weight because the selected file contains no population field. Reported statistics include the mean, median, standard deviation, coefficient of variation, percentile ratio, Gini coefficient, and Theil T. They describe the distribution across administrative regions, not across people. For additional geographic context, a one-way decomposition reports the share of total log-index sum of squares located between country means. This is descriptive variance accounting, not a causal country effect.

4.2 Spatial weights and global association

The prespecified outcome for spatial inference is $x_i = \log(\text{GDP index}_i)$. Logging makes proportional gaps more symmetric and reduces leverage from a few corporate and capital regions; the unlogged index is retained as a sensitivity check. The primary binary graph links regions whose generalized polygons share a boundary or point (first-order queen contiguity). Its 590 undirected edges are row standardized:

$$w_{ij}^R = \frac{w_{ij}}{\sum_j w_{ij}}.$$

The graph has 21 zero-neighbor regions and 25 components. Zero-neighbor regions remain in every descriptive table and map but are not assigned an artificial neighbor. Spatial inference uses the induced graph of the remaining 255 regions, re-centering the outcome in that sample.

For centered values $z_i = x_i - \bar{x}$, global Moran's statistic is

$$I = \frac{n}{S_0} \frac{\sum_i \sum_j w_{ij}^R z_i z_j}{\sum_i z_i^2}, \qquad S_0 = \sum_i \sum_j w_{ij}^R.$$

The null is random labeling of the observed values over the fixed graph. A one-sided greater test uses 99,999 permutations and the finite simulation correction

$$p = \frac{1 + \#\{I^{(b)} \geq I_{\text{obs}}\}}{B + 1}.$$

This evaluates positive spatial association, not sampling uncertainty in the Eurostat estimates.

4.3 Local association and robustness

For each region with at least one contiguity neighbor, Anselin's local statistic is proportional to $z_i \sum_j w_{ij}^R z_j$. Conditional randomization holds the focal z_i fixed and samples neighbor values without replacement from the other regions. Two-sided local p-values are centered on each region's permutation distribution. Quadrants are defined by the signs of z_i and its spatial lag: high-high (HH), low-low (LL), high-low (HL), and low-high (LH). "High" and "low" are relative to the spatial-analysis sample center, not necessarily to the EU benchmark of 100.

The main local result applies Benjamini-Hochberg adjustment at nominal $q = 0.05$ across all 255 tests. Holm-adjusted results provide a stricter family-wise-error check. Global sensitivity specifications use the raw index, a rank-normal transformation, rook contiguity, an EU27-only sample, the largest connected component, and a symmetric 5-nearest-neighbor graph on that component. Nearest-neighbor centroids and distances are calculated only after projection to EPSG:3035.

5 Results

5.1 Magnitude of regional disparities

Table 1: Region-equal descriptive statistics, 2023

Statistic	Value
Regions	276
Countries	31
Mean	89.6
Median	84.0
Standard deviation	36.7
Coefficient of variation	0.410
10th percentile	50.0
90th percentile	131.0
90/10 ratio	2.620
Gini coefficient	0.219
Theil T	0.078
Share below 75	39.1%
Share at or above 100	33.7%
Between-country share of log variance	54.2%

Notes: GDP per capita is an index with EU27 population-weighted average equal to 100. Shares and distributional statistics weight every region equally. The between-country entry is the share of total log-index sum of squares accounted for by deviations of country means.

Table 1 shows substantial dispersion. The index ranges from 26 in Şanlıurfa, Diyarbakır (TRC2) to 251 in Eastern and Midland (IE06). The median is 84, the standard deviation is 36.7, and the coefficient of variation is 0.410. The 90th-percentile region has 2.62 times the index of the 10th-percentile region. The Gini coefficient is 0.219 and Theil T is 0.078.

39.1% of regions fall below 75, while only 33.7% reach or exceed the EU benchmark. These are regional fractions, not population shares. Country means account for 54.2% of log-index variation, leaving a large within-country component as well. Figure 1 makes both structures visible: lower values are common in the east and southeast, but high capital or metropolitan regions appear within several lower-income countries, while important internal gradients are also visible in western member states.

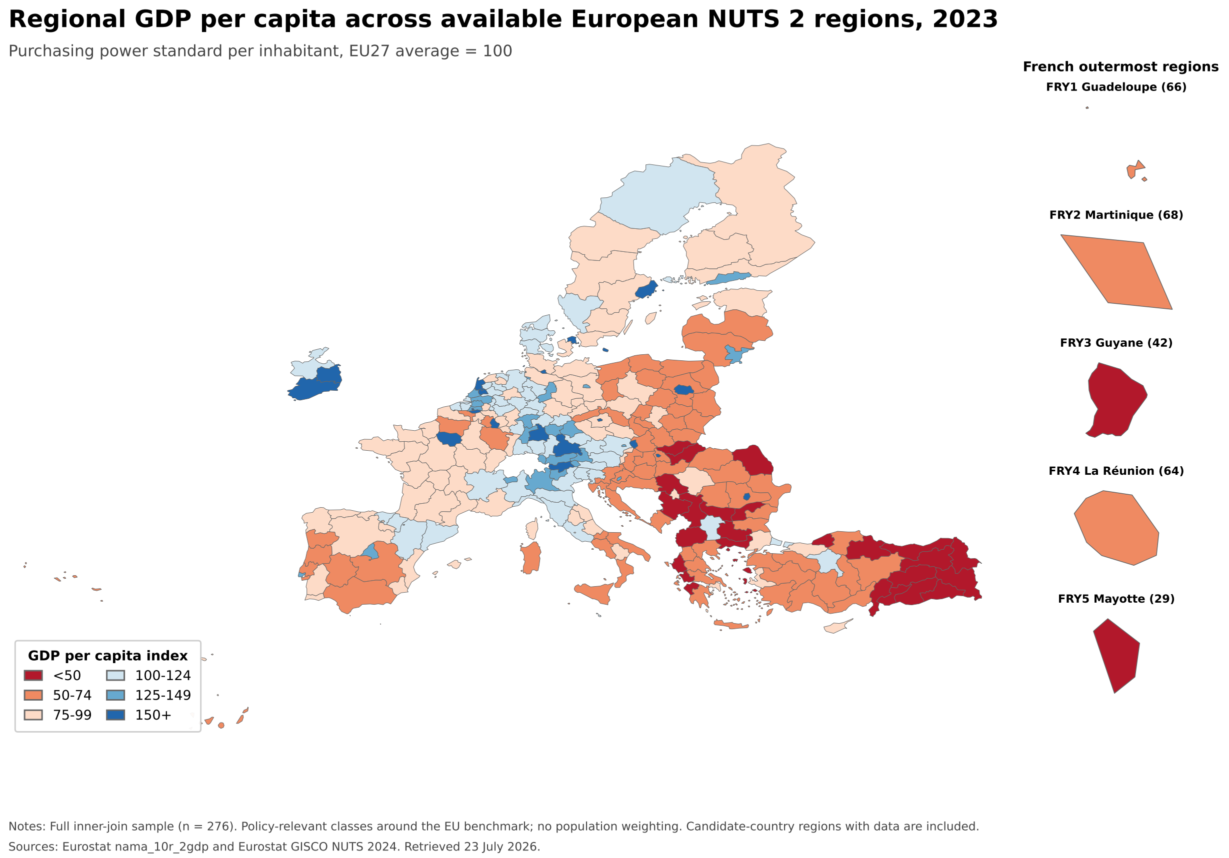

Figure 1: GDP per capita by NUTS 2 region

5.2 Global spatial association

Table 2: Global Moran results and sensitivity checks

Specification	n	Moran's I	$E[I]$	Permutation p
Primary: log, queen	255	0.600	-0.004	0.00001
Raw index, queen	255	0.460	-0.004	0.00001
Rank-normalized, queen	255	0.587	-0.004	0.00001
Log, rook	255	0.599	-0.004	0.00001
EU27 only: log, queen	223	0.503	-0.005	0.00001
Largest component: log, queen	238	0.600	-0.004	0.00001
Largest component: log, symmetric 5-NN	238	0.552	-0.004	0.00001

Notes: All specifications use 99,999 one-sided random-label permutations with fixed, recorded seeds. Queen and rook samples omit only zero-neighbor regions. The nearest-neighbor graph is symmetric and restricted to the largest component; its centroids use EPSG:3035.

The primary Moran estimate is 0.600, far above its random-label expectation of approximately -0.004, with the minimum attainable permutation p-value of 0.00001 (Table 2). A region with high log GDP per capita therefore tends to have high-valued polygon neighbors, and the same is true for low values. Moran's I is a measure of arrangement, not the magnitude of inequality summarized by the Gini or Theil index.

The conclusion is insensitive to reasonable choices. The raw-index estimate is 0.460, showing that the result is not created by logs. Rook contiguity gives 0.599. Restricting the sample to EU27 regions reduces the estimate to 0.503 but leaves a large positive pattern. Results for the largest component and its projected nearest-neighbor graph are likewise strongly positive. Figure 2 displays the global relationship.

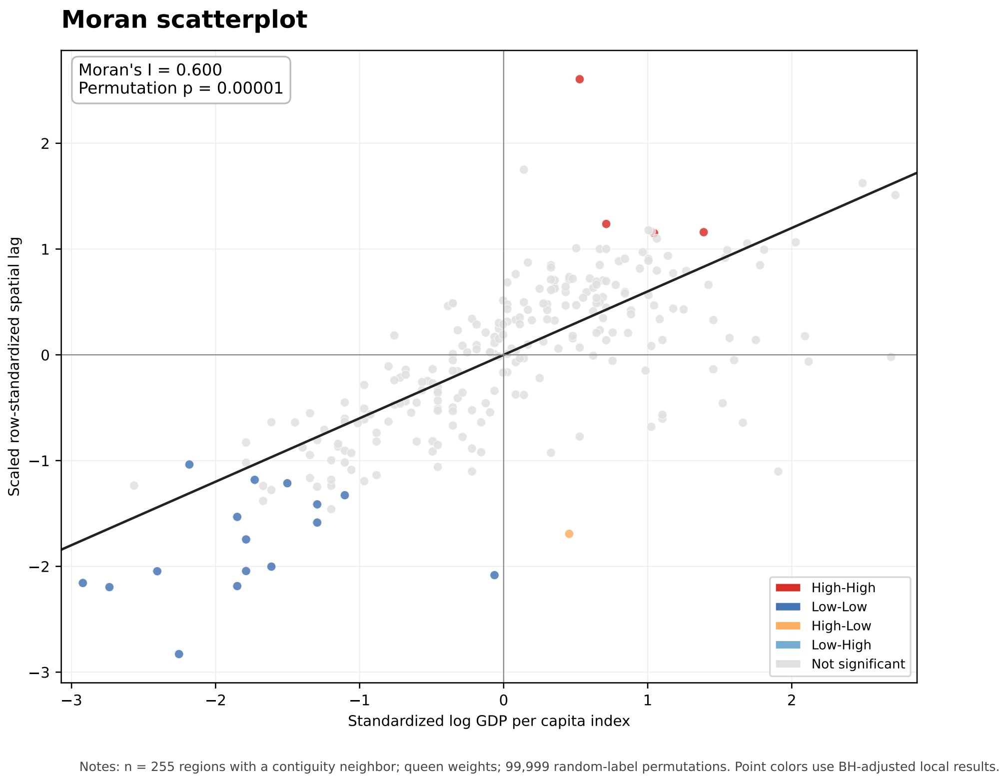

Figure 2: Moran scatterplot for the primary specification

5.3 Local clusters and spatial outliers

Table 3: Local Moran classifications after multiplicity adjustment

Category	BH $q \leq 0.05$	Holm
High-High	4	1
Low-Low	16	7
High-Low	1	1
Low-High	0	0
Not significant	234	246
No contiguity neighbor	21	21

Notes: Counts cover the full joined sample. The 21 zero-neighbor regions are shown separately and are not tested. The first result column applies Benjamini-Hochberg adjustment at $q = 0.05$; the second applies the dependence-robust Holm family-wise-error adjustment at 0.05.

Local inference is intentionally more conservative than the global test. Benjamini-Hochberg retains 21 regions: 4 HH, 16 LL, 1 HL, and 0 LH (Table 3). The HH cores are Salzburg and Tirol in Austria, Schwaben in Germany, and Northern and Western Ireland. The LL cores lie mainly across eastern and southeastern Türkiye, with additional regions in Serbia, Bulgaria, and northern Greece. Yugozapaden, which contains Sofia, is the single HL spatial outlier: its index is high relative to the analysis center while its contiguous neighbors have a low spatial lag. These are cluster cores under the chosen graph, not complete economic “clubs.”

Only 9 local results survive Holm adjustment. The contrast between pervasive global association and a short list of local survivors is not contradictory: global Moran aggregates weakly aligned edges across the entire graph, whereas each local test faces a stringent multiplicity burden. Figure 3 reports the BH-adjusted categories and explicitly marks regions without a contiguity neighbor.

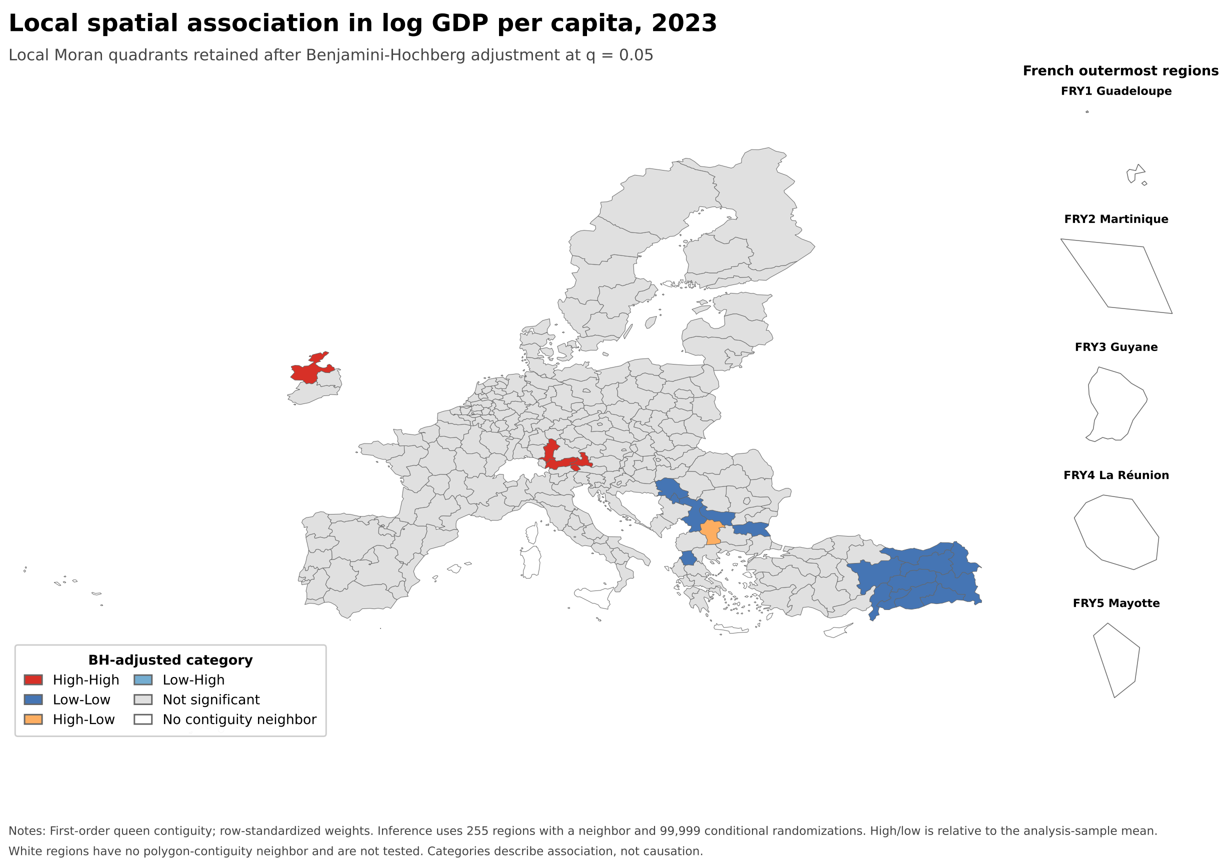

Figure 3: Benjamini-Hochberg-adjusted local Moran clusters

6 Limitations

Several limitations bound the interpretation.

- **Cross-section.** One year cannot establish beta convergence, sigma convergence, persistence, or a change in inequality.
- **No causal design.** Spatial association is compatible with many mechanisms, including market access, national institutions, industrial structure, sorting, common shocks, and shared history. Moran's I does not identify spillovers or policy effects.
- **Production versus welfare.** Regional GDP is workplace production, not household living standards. Commuting and multinational profit attribution can produce exceptional capital or corporate regions [Eurostat, 2025]. National rather than regional price converters may conceal subnational cost-of-living differences.
- **Coverage and revisions.** The available-data sample omits EFTA, the United Kingdom, and several Balkan regions. 101 observations are provisional and 1 is estimated. Eurostat serves the latest vintage, so future downloads may differ from the hashed snapshot.
- **Zoning and geometry.** Results are conditional on NUTS 2024 boundaries and 1:20 million generalized polygons. The modifiable areal unit problem and boundary generalization can affect both dispersion and adjacency. Twenty-one island, outermost, or otherwise geographically disconnected regions cannot enter the primary contiguity statistic.
- **Permutation null.** Random labeling answers whether this graph is more spatially organized than arbitrary reassignment. It is not a superpopulation sampling model and does not account

for measurement error.

- **Multiple testing.** Local Moran tests overlap spatially. Therefore, BH-adjusted values are reported at nominal $q = 0.05$ without claiming guaranteed false-discovery-rate control. Holm provides a dependence-robust family-wise-error check.

7 Conclusion

Regional GDP per capita in the available Eurostat NUTS 2 cross-section is both unequal and spatially organized. The region-equal Gini coefficient of 0.219 and 90-to-10 ratio of 2.62 summarize a wide distribution; country means explain just over half of log variation, leaving substantial within-country heterogeneity. Global Moran's I of 0.600 shows that this distribution is not randomly arranged over contiguous regions. The positive association persists in every prespecified sensitivity check.

Local evidence locates a small set of high-high cores in central Europe and Ireland and a larger low-low concentration in the southeast, but only 21 regions are retained after BH adjustment, and only 9 survive Holm adjustment. The appropriate conclusion is descriptive: Europe's regional production geography in 2023 exhibited strong spatial clustering. Explaining that geography requires longitudinal data and a research design capable of separating agglomeration, institutions, composition, and policy.

Data and code availability

All code, raw snapshots, hashes, metadata, anti-joins, processed data, spatial weights, tables, figures, and build instructions are stored with this paper. The analysis uses only official Eurostat and GISCO inputs. Re-running the download script may retrieve a revised vintage; reproducing the reported numbers requires the archived raw files and recorded checksums.

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